

Sai Chaudhry

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EDUCATION

Carnegie Mellon University

B.S. Machine Learning & Statistics

Pittsburgh, PA

May 2029

EXPERIENCE

Novig – Prediction Market Exchange

Quantitative Trading & Market-Making Developer

New York, NY

May 2026 – Present

- Contribute to the pricing, quoting, and market-making stack for a peer-to-peer prediction-market exchange that market-makes same-game parlays and prices customer RFQs (request-for-quote) in real time
- Built a first-principles same-game parlay pricing engine from scratch: a Gaussian-copula joint-probability model matching a paid vendor's correlation-adjusted prices to within $\pm 1\%$ across 1,000+ held-out games in four leagues (MLB, NBA, NHL, WNBA); generalized the 2-leg analytic form to an N-leg multivariate copula (nearest-PSD repair, Cholesky factorization, seeded Monte Carlo) and isolated a durable $\sim 3\text{--}4$ point edge vs. soft-book lines through adversarial walk-forward backtests
- Worked on cross-venue matching across Kalshi, Novig, and Polymarket — mapping and reconciling equivalent markets between systems so the market-making engine can source and hedge liquidity across books
- Assisted on the external market-making framework that lets a professional market-making firm stream two-sided liquidity into the exchange, contributing to the batch-quoting and mass-cancel (by event/market/outcome) paths over real-time order-book and tick market-data feeds
- Helped restore the live parlay RFQ quoting funnel and validated a per-leg multi-book fill on a 472-quote sample, recovering $\sim 50\%$ of previously dropped quote volume while holding $\sim 96\%$ seller-safety

Morgan Stanley – Analytics Team

Applied Quantitative Research

Sewickley, PA

May 2025 – Aug 2025

- Analyzed \$50M+ in portfolio allocations, performance trends, sector exposures, and risk metrics, translating complex data into actionable insights and client presentations for senior advisors
- Researched estate planning, trust structures, and wealth-transfer strategies for UHNW families and applied valuation analysis to 15+ companies, directly informing advisor investment recommendations

CMU – HCI Lab

Machine Learning Research Intern

Pittsburgh, PA

Sep 2025 – May 2026

- Built a smartwatch sensing pipeline streaming 8–10 hrs/day of IMU data to classify real-time hyperactivity states in children with ADHD, improving calm-moment detection precision from 40% to 80.6% across 9 families ($+2\times$ over baseline)
- Showed personalized models ($R^2 = .40$) outperformed global models ($R^2 = .23$), supporting human-in-the-loop behavioral sensing design

PROJECTS

TRADERS @ CMU | Prediction Markets Team

Jan 2026 – Present

- Engineered an automated arbitrage bot executing cross-platform prediction-market positions in real time, generating 250% ROI and a 12/13 record across NFL playoff markets by pricing outcomes ahead of line movement
- Used Bayesian updating with volume signals to fade public bias on mispriced Kalshi political contracts, capturing edge on probability dislocations

SKILLS

Languages: Python (PyTorch, NumPy, pandas), Rust, TypeScript, SQL, C/C++, Git

Trading & Market Making: Limit Order Books, Matching, Quoting, Mass Quote/Cancel, Kill Switch, RFQ Pricing & Execution, Market-Data Feeds, Position & Exposure Risk, Statistical Arbitrage

Quant & Modeling: Monte Carlo, Gaussian Copulas, Cholesky/PSD Repair, Bayesian Inference, Time-Series, Walk-Forward Backtesting, Prediction Markets (Kalshi, Novig, Polymarket)

Systems & Tools: Node.js, React, Microservices, REST/WebSockets, Docker, CI, Linux

Interests: Tennis, Basketball (NBA), Politics, Street Photography